ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2025 - 2026)

LEADER & ACHIEVER COURSE

PHASE : MLA,M,N,W,X,MAZC,D,E,O,P,U/V,MAAK,U/V,Y,Z,MAPB,MSP (SPARK PRO)

TARGET : PRE-MEDICAL 2026

Test Type : **MAJOR** Test Pattern : **NEET (UG)**

TEST DATE : 13-03-2026

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	2	4	3	2	4	2	1	4	3	1	4	2	1	4	3	4	1	3	1	4	3	3	3	2	3	1	2	3	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	1	2	2	4	1	1	2	2	3	4	2	3	1	4	1	2	2	4	1	2	2	1	1	1	4	3	3	2	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	1	3	4	2	3	3	2	4	1	1	3	3	1	4	1	3	4	1	1	3	1	4	2	3	1	4	1	2	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	3	2	2	3	3	1	1	2	4	4	1	3	2	2	1	4	1	3	1	2	2	4	3	4	2	3	3	4	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	2	4	4	3	1	3	3	2	1	4	3	2	3	2	4	4	1	3	2	2	1	3	2	4	1	3	2	4	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	4	2	3	1	1	3	4	3	4	3	2	2	3	1	2	3	3	4	4	4	2	1	3	1	1	4	1	3	1

HINT – SHEET

1. **Ans (2)**

Range $R = \frac{u^2 \sin 2\theta}{g}$ and same for q and $90 - \theta$

So same for 42° and 48°

Maximum height $H = \frac{u^2 \sin^2\theta}{2g}$

H is high for higher θ

So H for 48° is higher than H for 42°

Option (2)

2. **Ans (2)**

For speed up angle between $\vec{v}$ and $\vec{a}$ must be acute.

3. **Ans (4)**

$v_y \frac{d}{dt}(y) = \frac{d}{dt}(10t) - \frac{d}{dt}(t^2) = 10 - 2t$

At maximum height, $v_y = 0$

$10 - 2t = 0$

$2t = 10$

$t = 5$ sec.

$y = (10 \times 5 - 5 \times 5)_m = 25$ m

4. **Ans (3)**

Range are same

$T_1 = \frac{2u \sin\theta}{g}$

$T_2 = \frac{2u \sin(90 - \theta)}{g} = \frac{2u \cos\theta}{g}$

$T_1 T_2 = \frac{4u^2}{g^2} \sin\theta \cos\theta$

$\boxed{T_1 T_2 = \frac{2R}{g}}$

5. **Ans (2)**

Relative displacement between them will be zero at start and end of race.

$S_{rel} = U_{rel.} t + \frac{1}{2} a_{rel.} t^2$

$0 = 4t - \frac{1}{2} \times 1 \times t^2$

on solving $t = 8$

$xy = 16 \times 8 + \frac{1}{2} \times 1 \times 8^2$

$= 128 + 32 = 160$ m

6. Ans (4)

$S = 6 + 12t - 2t^2$

$V = 12 - 4t = 0$

$t = 3$

$S_{t=0} = 6$

$S_{t=3} = 6 + 36 - 18 = 24$

$S_{t=5} = 6 + 60 - 50 = 16$

distance = 26m

7. Ans (2)

$T = (2u \sin\theta) / g$

$= (2 \times 15 \times 3/5) / 10$

$= 18 / 10 = 1.8$ s

Since the man's height is negligible, he must reach the landing point at time T.

Minimum running speed:

v_{min} = distance / time

$= 9 / 1.8 = 5$ m/s

Final Answer: Option (2)

8. Ans (1)

$x = 1 + 2t^3 + 3t^4$

$v = 6t^2 + 12t^3$

v at $t_2 = 6 \times 4 + 12 \times 8$

$= 24 + 96 = 120$ m/s

v at $t_1 = 6 \times 1 + 12 \times 1$

$= 18$ m/s

$\Delta v = v_f - v_i = 120 - 18 = 102$ m/s

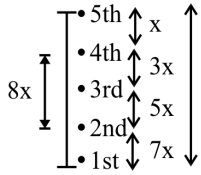
9. Ans (4)

$\therefore u = 0$

$y = 3x + 5x + 7x$

$\frac{y}{x} = \frac{15}{1}$

10. Ans (3)



$16x = 48$

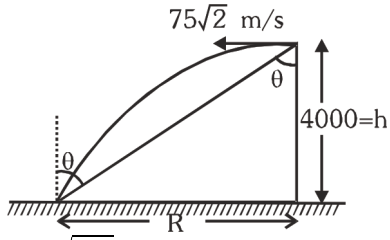
$x = \frac{48}{16} = 3$

$8x = 8 \times 3$

$= 24$ m

11. Ans (1)

Let t be the time taken by bomb to hit the target.



$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 4000}{10}} = 20\sqrt{2}$ sec

$R = ut = 75\sqrt{2} \times (20\sqrt{2}) = 3000$m

$\therefore \tan\theta = \frac{R}{h} = \frac{3000}{4000} = 37°$

12. Ans (4)

$V_A = \tan 53° = \frac{4}{3}$ m/s and

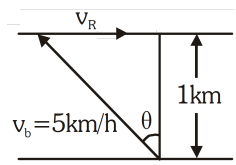
$V_B = \tan 37° = \frac{3}{4}$ m/s

$3 + \frac{3}{4}t_0 = \frac{4}{3}t_0$

$3 = \left(\frac{4}{3} - \frac{3}{4}\right) t_0$

$3 = \frac{7}{12}t_0 \Rightarrow t_0 = \frac{36}{7}$ sec

13. Ans (2)



$t = 15$ min $= \frac{15}{60} = \frac{1}{4}$ hr

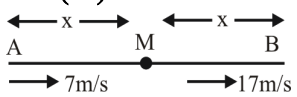
$t = \frac{1}{v_b \cos\theta} = \frac{1}{5\cos\theta} = \frac{1}{4}$

$\cos\theta = \frac{4}{5}$

$\sin\theta = \frac{3}{5}$

$\frac{3}{5} = \frac{v_R}{5} \Rightarrow v_R = 3\text{kmh}^{-1}$

14. Ans (1)



$v_m^2 = (7)^2 + 2 \times a \times x$

$(17)^2 = (7)^2 + 2 \times a \times 2x \quad ; v_m = 13$ m/s

$\frac{13-7}{a} = t_1 \; ; \; t_2 = \frac{17-13}{a}; \; \frac{t_1}{t_2} = \frac{6}{4} = \frac{3}{2}$

15. Ans (4)

(a) Scalar can have negative values.

(b) Scalar does not depend on orientation of axes

(c) A vector quantity must follow vector laws.

(d) If the resultant vector is zero then y component of resultant must be zero.

16. Ans (3)

$\tan\theta = 2t$

$\sec^2\theta \frac{d\theta}{dt} = 2;$ at $t = \frac{1}{2}$ sec $\Rightarrow \theta = 45°$

$\frac{d\theta}{dt} = \frac{2}{\sec^2 45°} = 1$

17. Ans (4)

$\text{Area} = 2\int_0^1 x^2 dx = 2\left[\frac{x^3}{3}\right]_0^1 = \frac{2}{3}$

18. Ans (1)

1 m'/s' = 1 × 10 m/20s

= 0.5 m/s

∴ 1 m/s = 2 m'/s'

1 watt' = 1 kg' m'^{2}/s'^{3}

$= 1 \times 5\,\text{kg} \times \frac{10^2 m^2}{20^3 s^3}$

$= \frac{1}{16}\text{kg}\,m^2/s^3$

∴ 16 watt$'$ = 1 watt

a & b are correct

19. Ans (3)

$m = (v)^a(d)^b(g)^k$

$M = (LT^{-1})^a(ML^{-3})^b(LT^{-2})^k$

$a - 3b + k = 0$

$-a - 2k = 0, b = 1$

$k = -3$

$|2k| = 6$

20. Ans (1)

$v = \lambda^a g^b \rho^c$

using dimension formula

$\Rightarrow [M^0L^1T^{-1}] = [L^1]^a [L^1T^{-2}]^b [M^1L^{-3}]^c$

$\Rightarrow [M^0L^1T^{-1}] = [M^c L^{a+b-3c} T^{-2b}]$

$\therefore c = 0, a + b - 3c = 1, -2b = -1 \Rightarrow b = \frac{1}{2}$

Now $a + b - 3c = 1$

$\Rightarrow a + \frac{1}{2} - 0 = 1$

$\Rightarrow a = \frac{1}{2}$

$\therefore a = \frac{1}{2}, b = \frac{1}{2}, c = 0$

21. Ans (4)

Terminal velocity of a spherical body in liquid

$\Rightarrow V_t \propto r^2$

$\Rightarrow \frac{\Delta V_t}{V_t} = 2 \cdot \frac{\Delta r}{r}$

$\Rightarrow \frac{\Delta V_t}{V_t} \times 100\,\% = 2\frac{(0.1)}{5} \times 100 = 4\,\%$

Also $V_t \propto r^2$

Reason R is false

Final Answer: A is true but R is false.

22. Ans (3)

Least count (LC) = $\left(\frac{b-a}{b}\right)$ MSD ;

here a = 9,

b = 10 ; 1 MSD = 1 mm

$\text{LC} = \left(\frac{10-9}{10}\right) 1\text{ mm} = 0.1\text{ mm}$

There is zero error in measurement.

As the zero of the vernier lies to the left of the main scale, it has negative zero error. For negative zero error, the coinciding division is to be read from right and it has 5th division coinciding from right.

Thus zero error = ZE = −5 × 0.1 mm = −0.5 mm = −0.05 cm

Reading at measurement : MSR = 8.5 cm

VSR = 8th VSD × 0.1 mm = 0.08 cm

Corrected reading = MSR+VSR × LC − zero error

= 8.5 cm + 0.08 cm − (−0.5mm) = 8.63 cm

23. Ans (3)

(A) $E = \frac{1}{2}LI^2 \Rightarrow [L] = \frac{[E]}{[I^2]}$

$\Rightarrow [L] = \frac{[M^1L^2T^{-2}]}{A^2} = [M^1L^2T^{-2}A^{-2}]$ (A → R)

(B) $E = \frac{1}{2}KT$

$\Rightarrow [K] = \frac{[E]}{[T]} = \frac{[M^1L^2T^{-2}]}{[K]} = [M^1L^2T^{-2}K^{-1}]$

(B ⟶ D)

(C) $\frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi d} \Rightarrow [\mu_0] = \frac{[F]}{[I^2]} = \frac{[M^1L^1T^{-2}]}{A^2}$

$[\mu_0] = [M^1L^1T^{-2}A^{-2}]$ (C ⟶ S)

(D) $R = \rho\frac{\ell}{A} = [\rho] = [RL] = \left[\frac{\rho}{I^2}.L\right]$

$\Rightarrow [\rho] =$

$\frac{[M^1L^2T^{-3}L]}{A^2} = [M^1L^3T^{-3}A^{-2}]$ (D → Q)

24. Ans (3)

For a particular vernier calliper, ZE cancels out when subtracting two measurements from same vernier calliper.

25. Ans (2)

$a = \frac{40}{8} = 5\ m/s^2$

$s = \frac{1}{2}at^2 = 10m$

$\therefore W_g = 40 \times 10 = 400$ J

26. Ans (3)

Mass = m

Initial linear momentum ⇒ $p_i = p$

Final linear momentum ⇒ $p_f = p_i + 200\%$ of p_i

$= p + \frac{200}{100} \times p$

$= p + 2p = 3p$

Initial kinetic energy

$\Rightarrow E_{ki} = \frac{p_i^2}{2m} = \frac{p^2}{2m} = E_K$(say)

Final kinetic energy

$\Rightarrow E_{ki} = \frac{p_i^2}{2m} = \frac{(3p)^2}{2m} = 9\left(\frac{p^2}{2m}\right) = 9E_K$

Absolute change in kinetic energy

$\Rightarrow \Delta E_K = (E_{Kf} - E_{Kf}) = (9E_K - E_K) = 8\ E_K$

Percentage change in kinetic energy

$\Rightarrow \frac{\Delta E_K}{E_{ki}} \times 100\% = \frac{8E_k}{E_k} \times 100\% = +800\%$

27. Ans (1)

Equilibrium is where the force F is zero.

Stability is determined by the sign of $\frac{dF}{dx}$:

Stable if $\frac{dF}{dx} < 0$.

$F = 2x^2 - 3x - 2$

For equilibrium,

$F = 0: 2x^2 - 3x - 2 = 0$

$\Rightarrow (2x + 1)(x - 2) = 0.$

$x = -\frac{1}{2}$ or $x = 2$.

$\frac{dF}{dx} = 4x - 3$

At $x = 2$: $\frac{dF}{dx} = 4(2) - 3$

$= 5 > 0$ (Unstable).

At $x = -\frac{1}{2}$: $\frac{dF}{dx} = 4\left(-\frac{1}{2}\right) - 3$

$= -2 - 3 = -5 < 0$ (Stable).

28. Ans (2)

$H = \frac{v^2}{2g}$

$W_g = -mgH = -mg\left(\frac{v^2}{2g}\right) = \frac{-1}{2}mv^2$

$= \frac{-1}{2}\left(\frac{1}{10}\right)(5)^2 = -1.25$ J

29. Ans (3)

Potential energy of spring $U = \frac{F^2}{2k}$

$\Rightarrow \frac{U_1}{U_2} = \frac{k_2}{k_1} = \frac{3000}{1500} = 2 : 1$

[If F = constant]

30. Ans (4)

Explain the question : In this question, we need to find how much mass of water (M) a 2 Kw pump can lift to a height of 10 m in 1 minute.

Concept : This question is based on power.

Solution : Power, $P = \frac{\text{work done}}{\text{time}} = \frac{MgH}{t}$

$\Rightarrow M = \frac{P \times t}{gH}$

Given here

P = 2kW = 2000 watt, t = 1 min = 60 sec

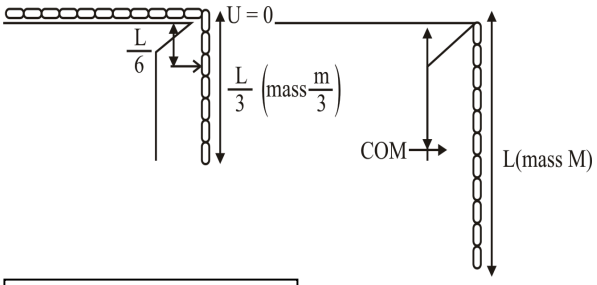
$g = 10\ m/s^2$, H = 10 M

$\Rightarrow M = \frac{2000 \times 60}{10 \times 10} = 1200$ kg ≅ 1200 litre

(1 litre ≅ 1 kg)

Final answer : Option (4)

31. Ans (2)



$$\boxed{pE_i = -\frac{mg}{3}\frac{L}{6} = -\frac{mgL}{18}}$$

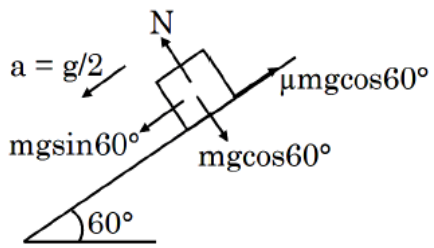
$$\boxed{pE_f = -mg\frac{L}{2}}$$

Now COME law

Gain in KE = Loss in PE

$$= \left(-\frac{mg\ell}{18}\right) - \left(\frac{-mg\ell}{2}\right) = \frac{8mg\ell}{18} = \frac{4mg\ell}{9}$$

32. Ans (1)



$mg \sin 60^o - \mu mg \cos 60^o = ma$

$g\sin 60 - \mu g \cos 60 = \frac{g}{2}$

$\frac{\sqrt{3}}{2} - \frac{\mu}{2} = \frac{1}{2}$

$\mu = \sqrt{3} - 1$

33. Ans (2)

Since, force needed to overcome frictional force.

34. Ans (2)

$$F = \frac{W}{\mu} = \frac{1 \times 9.8}{0.2} = 49N$$

35. Ans (4)

The sand bag is placed on a trolley that is moving with a uniform speed of 27 km/h. The external forces acting on the system of the sandbag and the trolley is zero. When the sand starts leaking from the bag, there will be no change in the velocity of the trolley. This is because the leaking action does not produce any external force on the system. This is in accordance with Newton's first law of motion. Hence, the speed of the trolley will remain 27 km/h.

36. Ans (1)

Weight of the disc will be balanced by the force applied by the bullet on the disc in vertically upward direction.

$F = 2nmv = Mg$

$= 2 \times 20 \times 0.05 \times 6 = M(10)$

$$M = \frac{2 \times 20 \times 0.05 \times 6}{10} = 1.2 \text{ kg}$$

37. Ans (1)

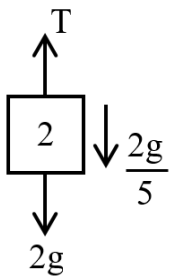
Due to friction ($a = \mu g$) velocity of block will become equal to velocity of belt. Relative motion between two will stop.

$v = at = \mu gt$

$= 0.2 \times 10 \times 4 = 8$ m/sec

38. Ans (2)

$$a = \frac{2g}{2+3} = \frac{2g}{5} \text{ m/s}^2$$



$$T = M\left(g - \frac{2g}{5}\right) = \frac{6g}{5}N$$

39. Ans (2)

$F_s = (2.5g \cos 37°) \sin 37°$

$= 25 \times \frac{3}{5} \times \frac{4}{5} = 12N$

40. Ans (3)

Force produced by the engine $F = \frac{P}{v} = \frac{30 \times 10^3}{30}$

$= 10^3 N$

$$\text{Acceleration} = \frac{\text{Forward force by engine-resistive force}}{\text{mass of car}}$$

$$= \frac{1000 - 750}{1250} = \frac{250}{1250} = \frac{1}{5} \text{m/s}^2$$

41. Ans (4)

$$W = -\int F\,dx$$

$$W = -\int_{20}^{30} 0.1x\,dx$$

$$W = -0.1\left[\frac{x^2}{2}\right]_{20}^{30}$$

$$W = -0.1\left[\frac{900-400}{2}\right] = -25\text{ J}$$

From work energy theorem $W = K_f - K_i$

$$\Rightarrow \quad -25 = K_f - \frac{1}{2}10(10)^2$$

$$\Rightarrow \quad K_f = 475\text{ J}$$

42. Ans (2)

$\because\ U = \sqrt{2gH} = \sqrt{32\times 2g} = \sqrt{64g}$

$\because\ t = \dfrac{U}{g} = \dfrac{\sqrt{64g}}{g}$

$\therefore\ 4 = \sqrt{\dfrac{64}{g}} \quad \therefore g = 4\text{ m/s}^2$

$\because\ S_{n^{th}} = U + \dfrac{a}{2}(2n-1)$

$\therefore\ S_{4^{th}} = \sqrt{64\times 4} - \dfrac{4}{2}(2\times 4 - 1)$

$= 16 - 14 = 2$ m

43. Ans (3)

Under area of $F - t = \Delta P$

so, P will be maximum at t = 15 sec.

44. Ans (1)

$$\frac{dw}{dt} = P = 2x$$

$FV = 2x$

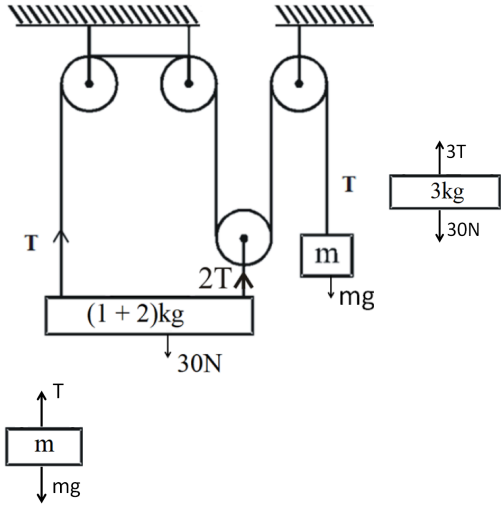
$$\left(mv\frac{dv}{dx}\right)(v) = 2x$$

$$\int_0^v v^2\,dv = \left(\frac{2}{m}\right)\int_0^x x\,dx$$

$$\frac{v^3}{3} = \frac{2}{m}\frac{x^2}{2}$$

$$v = \left(\frac{3}{m}x^2\right)^{1/3}$$

45. Ans (4)



3T = 30 .... (1)

T = mg .... (2)

Put T in equation ---------- (1)

3mg = 30

m = 1kg

46. Ans (1)

(L.R.)

$$3A + 2B \longrightarrow C + \frac{1}{2}D$$

12 mol 12 mol

Moles of D = $\dfrac{1}{6}\times 12 = 2$

$\dfrac{60}{M_w} = 2 \Rightarrow M_w = 30$

47. Ans (2)

NCERT Pg # 25

$C_2H_6 + \dfrac{7}{2}O_2 \rightarrow 2CO_2 + 3H_2O(g)$

moles of $C_2H_6 = \dfrac{3}{30} = 0.1$

moles of $H_2O(g) = 0.1 \times 3 = 0.3$

vol of $H_2O(g)$ at STP = 0.3 × 22.4 = 6.72 L

48. Ans (2)

Let the mass of pure $CaCO_3$ in the sample be x g

According to the reaction

100 g $CaCO_3 \longrightarrow$ 56 g CaO

So,

(56/100) × x = 56

⇒ x = 100g

% purity = (100/120) × 100 = 83.3%

Final Answer:

Percentage purity of $CaCO_3$, = 83.3%

Correct Option: (2) 83.3%

49. Ans (4)

Number of molecules = $1.4 \times N_A$

$= 1.4 \times 6.022 \times 10^{23}$

$= 8.43 \times 10^{23}$

Total atoms = $8.43 \times 10^{23} \times 4$

$= 3.372 \times 10^{24}$

Final Answer :

Option (4)

50. Ans (1)

Concept: Mole relation: 1 mole of CO_2 has 2 O atoms; 1 mole of CO has 1 O atom.

Solution:

Molar mass of CO_2 = 12 + (2×16)

= 44 g/mol

Moles of CO_2 in 22g:

$n(CO_2) = 22/44 = 0.5$ mol

Moles of O atoms in $CO_2 = 0.5 \times 2$

= 1 mol O atoms

Each CO molecule has 1 O atom ⟶ need 1 mol CO

Mass of 1 mol CO = 12 + 16 = 28 g

Final answer: option (1)

51. Ans (2)

$$\text{Equivalent mass of metal} = \frac{\text{mass of metal}}{\text{mass of oxygen}} \times 8$$

$$= \frac{68}{32} \times 8 = 17$$

52. Ans (2)

$$\% = \frac{\text{Atomic weight} \times (\text{no. of atoms} = 1)}{\text{minimum molecular weight}} \times 100$$

$$0.4 = \frac{39 \times 1}{(M_w)_{min.}} \times 100$$

$$(M_w)_{min} = \frac{39}{0.40} \times 100 = 9750$$

53. Ans (1)

$C_{\frac{38.8}{12}} H_{\frac{16}{1}} N_{\frac{45.2}{14}} \Rightarrow C_{3.23}H_{16}N_{3.23}$

$= C_1H_5N$

$= CH_3NH_2$

54. Ans (1)

no. of moles of Co_2 left = $n_2 - n_1$

$$= \frac{200 \times 10^{-3}}{44} - \frac{10^{21}}{6.02 \times 10^{23}}$$

$= 2.88 \times 10^{-3}$ moles

Final Answer - option (1)

Level - Medium

55. Ans (1)

$H_3PO_4 + 3NaOH \longrightarrow Na_3PO_4 + 3H_2O$

H_3PO_4 donates **3 H^+ ions**

Valency factor of H_3PO_4 = 3

Molecular weight of H_3PO_4 = (3×1) + 31 + (4×16)

= 98

Equivalent weight = 98 ÷ 3 = 98/3

FINAL ANSWER: Option (1)

56. Ans (4)

$$\underset{+6}{\underline{Cr}_2O_7^{2-}} + \underset{+4}{\underline{S}O_3^{2-}} \rightarrow \underset{+3}{\underline{Cr}^{3+}} + \underset{+6}{\underline{S}O_4^{2-}}$$

(6) ⟷ (2)

$$8H^+ + Cr_2O_7^{2-} + 3SO_3^{2-} \longrightarrow 2Cr^{3+} + 3SO_4^{2-} + 4H_2O$$

no. of Cr^{3+} : no. of H^+ = 2 : 8

= 1 : 4

Final answer: Option (4)

57. Ans (3)

Oxidation number of S in H_2SO_4 is +6 which is maximum. hence it can not further oxidise and can not behave as reducing agent.

58. Ans (3)

Oxidising agent has a tendency to gain electrons

59. Ans (2)

Question Explanation: To Calculate equivalent weight of H_3PO_2 disproportionation reaction.

Concept: Equivalent weight in a redox reaction.

Solution:

$$\overset{+1}{H_3PO_2} \longrightarrow \overset{-3}{PH_3} + \overset{+3}{H_3PO_3}$$

H_3PO_2 disproportionate , So v.f while getting oxidised = 3 – 1 = 2

v.f while getting reduced = 1 – (–3) = 4

Overall v.f (n)

$$\frac{1}{n} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$n = \frac{4}{3}$$

$$\text{Wet} = \frac{M \times 3}{4} = \frac{3M}{4}$$

Final Answer: Option (2)

60. Ans (3)

Question Explanation:

To calculate mass of $K_2Cr_2O_7$, required to oxidise Mohr's salt.

Concept:

Low of chemical equivalence.

Solution:

$$\overset{+2}{F}eSO_4(NH_4)SO_4.6H_2O + K_2K_2\overset{+6}{Cr_2}O_7 \longrightarrow F_e^{3+} + Cr^{3+}$$

v.f of Mohr's salt = 1 (iron changing from + 2 to + 3)

v.f of $K_2Cr_2O_7 = 3 \times 2 = 6$

eq. of Mohr's salt = Gm eq. of $K_2Cr_2O_7$

$$\frac{750}{1000}L \times 0.6 \times 1 = \text{moles} \times 6$$

$$\Rightarrow \text{moles of } K_2Cr_2O_7 = \frac{750 \times 0.6}{1000 \times 6}$$

$$\text{wt of } K_2Cr_2O_7 = \frac{750 \times 0.6}{1000 \times 6} \times [39 \times 2 + 52 \times 2 + 16 \times 7]$$

$$= \frac{750 \times 0.6}{1000 \times 6} \times 294 = 22.05g$$

Final Answer:

Option (3)

61. Ans (4)

$$N_2H_4 \longrightarrow 2NO_3^-$$

$$6H_2O + N_2H_4 \longrightarrow 2NO_3^- + 16H^+ + 14e^-$$

62. Ans (1)

$$K = \frac{(0.4)^2}{(0.1)} \Rightarrow 1.6\ M$$

63. Ans (3)

$$K_C = \frac{K_f}{K_b} = \frac{K_f}{2500\ K_f} = \frac{1}{2500}$$

$$K_p = K_C(RT)^{\Delta n_g} = \frac{1}{2500}(0.0831 \times 1000)^1$$

$$K_p = 0.033$$

64. Ans (4)

	PCl_5 (g)	$\rightleftharpoons$ PCL_3 (g)	+ Cl_2 (g)
t = 0	2	–	–
t_{eq}	2 – x	x	x

$$\%a = 35\%\ ;\ \frac{35}{100} = \frac{x}{2} \Rightarrow x = 0.7$$

Total number of moles at equilibrium = 2 – x + x + x

= 2 + x

= 2 + 0.7 = 2.7

Final answer: 4

65. Ans (2)

Explain the question

Analyze ice-water equilibrium shifts.

Concept

Le Chatelier's Principle.

Solution

For the equilibrium $H_2O_{(s)} \rightleftharpoons H_2O_{(l)}$, the forward process (melting) is endothermic ($\Delta H>0$); thus, increasing temperature shifts the equilibrium forward. Regarding pressure, water exhibits anomalous behavior where the density of ice is less than liquid water, meaning the volume of ice is greater than the volume of water ($V_{ice} > V_{water}$). According to Le Chatelier's Principle, increasing pressure shifts the equilibrium toward the side with lower volume (the liquid phase). While the Assertion is true because both factors favor water formation, the Reason is false because it incorrectly claims the volume of water is greater than the volume of ice.

Final answer : Option 2

66. Ans (3)

Explain the Question : We need to find relation between two equilibrium constants for given two reactions.

Concept : This question is based on Factors affecting K_{eq}.

Solution :

$2A + B \rightleftharpoons 2C$; $K_{eq} = 40$

$\downarrow$ Reaction is reversed

$2C \rightleftharpoons 2A + B$; $K_{eq} = \frac{1}{40}$

$\downarrow$ Multiplied by $\frac{1}{2}$

$C \rightleftharpoons A + \frac{1}{2}B$; $K_{eq} = \left(\frac{1}{40}\right)^{\frac{1}{2}}$

$Keq = \sqrt{\frac{1}{40}}$

Final answer : 3

67. Ans (3)

Explain the question : We need to find direction of reaction if value of K_C and concentration of reactants and product are given.

Concept : This question is based on reaction quotient.

Solution :

$2A \rightleftharpoons B + C$; $K_c = 4 \times 10^{-3}$

At a given time $\Rightarrow [A] = [B] = [C] = 2 \times 10^{-3}$ M

$Q_c = \frac{[B][C]}{[A]^2}$

$Q_c = 1$

as $Q_c > K_c$

Reaction has a tendency to go in backward direction.

Final answer : 3

68. Ans (2)

Explain the Question : We need to find relation between K_P and total pressure (P) for reaction if % α is given.

Concept : This question is based on K_P involving degree of dissociation.

Solution : $PCl_5\,(g) \rightleftharpoons PCl_3\,(g) + Cl_2\,(g)$

t = 0	1	–	–
t_{eq} = 0.5	1 – 0.5	0.5	0.5

$(P)_{PCl_5} = X_{PCl_5} \cdot P_T = \frac{0.5P}{1.5} = \frac{P}{3}$

$(P)_{PCl_3} = X_{PCl_3} \cdot P_T = \frac{0.5P}{1.5} = \frac{P}{3}$

$(P)_{Cl_2} = X_{Cl_2} \cdot P_T = \frac{0.5P}{1.5} = \frac{P}{3}$

$K_P = \frac{\left(\frac{P}{3}\right)^2}{\left(\frac{P}{3}\right)} \Rightarrow K_P = \frac{P}{3} \Rightarrow P = 3K_P.$

Final answer : 2

69. Ans (4)

All the measurable properties of the system are variable at equilibrium state:

Incorrect: At equilibrium, all measurable properties are constant. They do not change with time.

Conclusion:

The statement that is not true is "All the measurable properties of the system are variable at equilibrium state."

Hence,

Option (4) is correct

70. Ans (1)

• The reaction is exothermic because heat (242 kcal) is a product. To favour the forward reaction, the temperature must be lowered.

• **Pressure**: Increasing pressure favours the side with fewer gas moles.

• Since the product side has fewer moles of gas, high pressure will favour product formation.

The correct conditions are low temperature and high pressure

Final Answer: Option 1.

71. Ans (1)

The formula for the degree of hydrolysis (h) for a WAWB salt is:

$$h = \sqrt{\frac{K_\omega}{K_a . K_b}}$$

- In this formula, three is no term for concentration (C).
- Therefore, whether the concentration is 0.001 M or 0.002 M, the value of h (represented here as x) remains the same.
- Result: $x_1 = x_2$.

Final Answer: Option (1)

72. Ans (3)

$N_1V_1 = N_2V_2$

$0.2 \times 2 \times V = 0.1 \times 2 \times 5$

$V = \frac{5}{2} = 2.5$ L

73. Ans (3)

Q. Explanation : We need to find highest degree of hydrolysis(h).

Concept : Degree of hydrolysis.

Explanation :

$h = \sqrt{\frac{Kw}{C \times Kb}}$

As concentration decreases, the degree of hydrolysis increases.

0.001 M NH_4Cl ⟶ Lowest concentration = Highest hydrolysis

Answer: option 3, 0.001 M NH_4Cl

74. Ans (1)

Explain the Question : We are asked why an aqueous solution of sodium carbonate (Na_2CO_3) is alkaline.

Concept : This question is based on hydrolysis of WASB

Solution :

Na_2CO_3 is a salt of a strong base (NaOH) and a weak acid (H_2CO_3).

In water, Na^+ ions do not hydrolyze (they come from a strong base).

CO_3^{2-} ions undergo hydrolysis :

$CO_3^{2-} + H_2O \rightleftharpoons 2OH^- + H_2CO_3$

This reaction produces OH^- ions, making the solution alkaline.

Final answer : 1

75. Ans (4)

Explain the Question : Need to find pH of the solution buffer solution .

Concept : This question is based on buffer solution.

Solution :

$\left.\begin{matrix} NH_4Cl & + & NH_4OH \\ \text{salt} & & \text{weak base} \end{matrix}\right\}$ make basic buffer.

$pOH = pKb + \log \frac{[salt]}{[base]}$

$pOH = 5 + \log \left(\frac{10^{-1}}{10^{-2}}\right)$

$pOH = 5 + \log 10$

$pOH = 5 + 1 = 6$

$pH = 14 - 6 = 8$

Final answer : 4

76. Ans (1)

Case-I

$NaOH + NH_4Cl \longrightarrow NaCl + NH_4OH$

1 mol	1 mol	0	0
0	0	1 mol	1 mol

(Neutral)

$[OH^-]_{NH_4OH} = \sqrt{K_b \times C}$

$(pOH)_I = \frac{1}{2} pK_b \left(C = \frac{1 \text{ mol}}{1L} = 1\right)$

Case II

$NH_4OH + NH_4Cl \Rightarrow$ Basic Buffer

1 mol 1 mol

$(pOH)_{II} = pK_b + \log \frac{[NH_4Cl] = 1}{[NH_4OH] = 1} = pK_b$

$\therefore \frac{(pOH)_I}{(pOH)_{II}} = \frac{1/2\ pK_b}{pK_b} = \frac{1}{2}$

77. Ans (3)

Concept : This question is based on

Precipitation occurs when : $K_{sp} = [Pb^{2+}][OH^-]^2$

Relation between pH and $[OH^-]$: pH + pOH = 14, $[OH^-] = 10^{-pOH}$

Solution :

$K_{sp} = [Pb^{2+}][OH^-]^2$

$1.2 \times 10^{-5} = 0.12 \times [OH^-]^2$

$[OH^-]^2 = \frac{1.2 \times 10^{-5}}{0.12} = 1.0 \times 10^{-4}$

$[OH^-] = 10^{-2}$ M

pOH = 2 ⇒ pH = 14 – 2 = 12

Final answer : 3

78. Ans (4)

Concept : This question is based on $\Delta S = \sum S_{products} - \sum S_{reactants}$

Solution :

Entropy of products = 2 × S(HCl)

$= 2 \times 186.7 = 373.4$ J K^{-1} mol^{-1}

Entropy of reactions = $S(H_2) + S(Cl_2)$

$= 130.6 + 223 = 353.6$ J K^{-1} mol^{-1}

$\Delta S = 373.4 - 353.6 = +19.8$ J K^{-1} mol^{-1}

Final answer : 4

79. Ans (1)

Question Explanation :

Question is asking about Match the sign of ΔH and ΔS to the nature of the reaction.

Concept :

This question is based on spontaneity of reaction.

Solution :

(P) Spontaneous only at low temperature :-

ΔH = –ve, ΔS = –ve, ΔG = –ve (A–P)

(Q) Spontaneous only at high temperature :-

ΔH = +ve, ΔS = +ve, ΔG = –ve (C–Q)

(R) Spontaneous only at all temperature :-

ΔH = –ve, ΔS = +ve, ΔG = –ve (B–R)

(S) Spontaneous only at all temperature :-

ΔH = +ve, ΔS = –ve, ΔG = +ve (D–S)

Final Answer :

Option (1)

80. Ans (1)

Concept:

- **Adiabatic Process:** An adiabatic process is one in which no heat is exchanged between the system and the surroundings (q = 0).
- **Reversible Process:** A reversible process is one that can be reversed by an infinitesimal change in a property of the system.

Mathematical Calculation:

For a reversible adiabatic process, the change in entropy (ΔS) is always zero. This is because there is no heat transfer (q = 0) and the process is reversible.

ΔS = q(rev)/T. Since q = 0, then ΔS = 0.

Final Answer:

The change in entropy (ΔS) is zero.

Therefore, the correct answer is option 1: Zero.

81. Ans (3)

First Law of Thermodynamics:

ΔU = q + w

0 = 208 J + w

w = –208 J

Final Answer

The values are q = +208 J and w = –208 J.

Therefore, the correct answer is (3)

q = +208 J, w = –208 J.

82. Ans (1)

Combustion reaction :-

$CH_4(g) + 2CO_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$

∵ Δng = 1 – 3 = 2

$\Delta H = -800 + \left(-2 \times \frac{8.314 \times 298}{1000}\right)$

ΔH = –805 KJ

Final Answer :

Option (1)

83. Ans (4)

$C + O_2 \longrightarrow CO_2$; $\Delta_C H = -390$ kJ mol^{-1}

1 mole = 44 g $\longrightarrow$ 390 kJ released

For 35 g $\longrightarrow \frac{390}{44} \times 35 = 310$ kJ

84. Ans (2)

For 1 mole HCl and NaOH; Heat = -57.3 kJ

For 0.25 mole; Heat = -57.3×0.25

$= -14.32$ kJ

(Unit should not be kJ mol^{-1})

85. Ans (3)

$\Delta H^o_{comb}(H_2) = \Delta H^o_f(H_2O)$

Calorific value = $\frac{\Delta H^o_{comb}(H_2)}{Mw_{H_2}}$

$-143 = \frac{\Delta H^o_f(H_2O)}{2}$

$\Delta H^o_f(H_2O) = -286$ kJ mol^{-1}

86. Ans (1)

Target equation = (eq. (ii) × 2) + eq. (iii) − eq.(i)

$\Delta H = (-393.5 \times 2) + (-285.8) - (-1299.5)$

$\Delta H = +226.7$ kJ

87. Ans (4)

$\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightarrow NO(g)$

Desired equation = $\frac{1}{2}x(1) - \frac{1}{2}x(2)$

$\Delta H = \frac{1}{2}(x - y)$

88. Ans (1)

NCERT-XI, Pg. # 172

$\frac{1}{2}H_{2(g)} + \frac{1}{2}X_{2(g)} \rightarrow HX_{(g)}\ \Delta H = -50\,kJ$

$\Delta H = \Sigma(B.E.)_{Reactants} - \Sigma(B.E.)_{Products}$

89. Ans (2)

$C_7H_{16} + 11O_2 \longrightarrow 7CO_2 + 8H_2O$

$\Delta H^0 = 7(-394) + 8(-286) - (-200)$

$\Delta H^0 = -2758 - 2288 + 200 = -4846$

90. Ans (3)

Question Explanation

To compare Q_1 and Q_2

Concept:

Laws of thermochemistry

Solution:

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O + Q_1$

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell) + Q_2$

liquidis more stable and have more force of attraction so in case of $H_2O(\ell)$ formation more heat is released .

So, $Q_2 > Q_1$

Final answer: Option (3)

91. Ans (2)

NCERT XI Page No. # 1.31/9

92. Ans (3)

Module

The correct answer is C & D

93. Ans (2)

NCERT XI Page No. # 8

94. Ans (2)

NCERT XI, Page No. # 6,7

95. Ans (3)

Module

96. Ans (3)

NCERT-XI, Pg. # 11

97. Ans (1)

NCERT-XI, Pg. # 10

98. Ans (1)

NCERT-XI, Pg. # 90, 91

99. Ans (2)
NCERT-XI, Pg. # 14

100. Ans (4)
NCERT XI Pg. # 13

101. Ans (4)
NCERT XI Pg. # 19,20

102. Ans (1)
NCERT Pg. No. 20, 21, 22, 2.2 - 2.2.1, 2.2.2, 2.2.3, 2.2.4, 2.2.5

103. Ans (3)
NCERT-XI, Pg. # 14, 15

104. Ans (2)
NCERT -XI, Pg. # 14, 15
Statements A, B, D and E are correct.

105. Ans (2)
NCERT-XI, Pg. # 14, 15

106. Ans (1)
NCERT XI, Pg.#23, 24

107. Ans (4)
NCERT Pg. # 18

108. Ans (1)
NCERT-XI, Pg. # 17-18

109. Ans (3)
NCERT-XI, Pg. # 16, 17

110. Ans (1)
NCERT-XI, Pg. # 16

111. Ans (2)
XI NCERT Pg. No. # 24

112. Ans (2)
NCERT-XI, Pg. # 19, 20, 21

113. Ans (4)
NCERT-XI, Pg. # 19, 20

114. Ans (3)
NCERT-XI, Pg. # 20

115. Ans (4)
NCERT-XI, Pg. # 21

116. Ans (2)
NCERT-XI, Pg. # 12 to 18

117. Ans (3)
NCERT-XI, Pg. # 20-23, 27

118. Ans (3)
NCERT-XI, Pg. # 19, 20, 21

119. Ans (4)
NCERT-XI, Pg. # 26, 27, 28

120. Ans (1)
NCERT-XI, Pg. # 27, 28
Pheophyceae (Brown algae) :-
Fucus, Dictyota, Sargassum
Rhodophyceae :-
Gelidium, Porphyra, Polysiphonia, Chondrus

121. Ans (1)
NCERT-XI, Pg. # 30

122. Ans (2)
NCERT Pg. No. 42

123. Ans (4)
NCERT Pg # 34

124. Ans (4)
NCERT-XI, Pg. # 29

125. Ans (3)
NCERT No. Pg.#29

126. Ans (1)
NCERT-XI, Pg. # 32

127. Ans (3)
NCERT-XI, Pg. # 29, 31, 33, 34

128. Ans (3)
NCERT-XI, Pg. # 32

129. Ans (2)
NCERT-XI, Pg. # 33

130. Ans (1)
NCERT-XI, Pg. # 31

131. Ans (4)
NCERT XI Pg.#33,34

132. Ans (3)
NCERT-XI, Pg. # 32, 33, 34
Selaginella, Cycas, Pinus, Eucalyptus, Salvinia

133. Ans (2)
NCERT-XI, Pg. # 26

134. Ans (3)
NCERT-XI, Pg. # 32

135. Ans (2)
NCERT-XI, Pg. # 26 ; Table 3.1

136. Ans (4)
NCERT Pg. # 43

137. Ans (4)
NCERT Pg. # 45

138. Ans (1)
NCERT Pg. # 45

139. Ans (3)
They are covered by ctenoid/cycloid scales.

140. Ans (2)
Pg. No. # 45

141. Ans (2)
Bath sponge is common name of Euspongia.

142. Ans (1)
NCERT, Pg. # 45-47

143. Ans (3)
NCERT–XI, Pg. # 56 (Fig. 4.18)

144. Ans (2)
NCERT-XI, Pg. # 47-48

145. Ans (4)
NCERT Pg.#42-45.

146. Ans (1)
NCERT-XII, Pg. # 104

147. Ans (3)
NCERT-XI, Pg. # 45, Para-4.2.9

148. Ans (2)
NCERT, Pg. # 47-48

149. Ans (4)
NCERT (XIth) Pg. # 53, IInd - Para

150. Ans (4)
NCERT XI Pg # 58

151. Ans (2)
NCERT XI Pg.#55

152. Ans (4)
NCERT, Pg # 47-48

153. Ans (2)
NCERT, Pg. # 39

154. Ans (3)
Module

155. Ans (1)
Module

156. Ans (1)
Question Explanation :
The question is asking to correctly match type of connective tissues in Column I with their respective characteristics or cell types in Column II.
Concept:
This question is based on connective tissue.
Solution:
A – Areolar tissue ⟶ Loose connective tissue
B – Adipose tissue ⟶ Contains fat cells
C – Ligaments ⟶ Mode of dense regular connective tissue
D – Bone ⟶ Contains osteocytes.
Final Answer: Option (1)

157. Ans (3)
Module

158. Ans (4)
Module

159. Ans (3)
NCERT XI Pg # 102, Para 7.1.1

160. Ans (4)
NCERT Page No. 101,102

161. Ans (3)
NCERT XI Pg. 111

162. Ans (2)
A-Ocellus, B-Compounds eye, C-Mandible, D-Maxilla, E-Labrum, F-Labium

163. Ans (2)
NCERT Page - 111 and 112

164. Ans (3)
NCERT Page - 112 and 114

165. Ans (1)
NCERT Pg.#112 (E), 113 (H)

166. Ans (2)
NCERT Pg.#114 (E), 114 (H)

167. Ans (3)
NCERT 11th Pg. No. 112

168. Ans (3)
Old NCERT, Pg. # 112

169. Ans (4)
NCERT Pg. # 114

170. Ans (4)
NCERT, Pg. # 82

171. Ans (4)
NCERT Pg. No.116/7.5.1

172. Ans (2)
NCERT Pg. No.119/7.5.2

173. Ans (1)
NCERT Pg. No.119/7.5.2

174. Ans (3)
NCERT Page - 118 and 119

175. Ans (1)
NCERT Pg. # 116

176. Ans (1)
NCERT-XI, Pg. # 118

177. Ans (4)
NCERT, Pg. # 81-82

178. Ans (1)
In male frogs, the ureters serve a dual function as urinogenital ducts. They carry both urine from the kidneys and sperm from the testes to the cloaca, where it is expelled from the body.ans-1

179. Ans (3)
NCERT, Pg. # 83

180. Ans (1)
NCERT Page - 116 to 119